PROJECT REPORT OF CO-OP Project AT Industry (Module-2)

ON

Mental Health Assessment Using Sentiment Analysis

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Anish Singal, Simranpreet Kaur

2210990116, 2210990854

Supervised By:

Dr. Shikha Tuteja

Associate Professor

Chitkara University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY

CHANDIGARH-PATIALA NATIONAL HIGHWAY, RAJPURA, PUNJAB, INDIA

CHITKARA UNIVERSITY

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"Mental Health Assessment using Sentiment Analysis"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of Dr. Shikha Tuteja. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Anish Singal, Simranpreet Kaur**

Date: 2210990116, 2210990854

This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

Dr. Shikha Tuteja

Associate Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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Anish Singal
ID No-2210990116

Simranpreet Kaur
ID No- 2210990854

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Abstract-This research paper addresses a major worldwide concern: the rising incidence of mental health illnesses, which is frequently made worse by the lack of clinical services and the stigma associated with conventional diagnosis. The system leverages social media as a "digital footprint" for psychological evaluation in order to meet the pressing need for scalable, non-invasive screening methods. To automatically detect distress indicators in unstructured text, we provide an advanced sentiment analysis method. Our technique uses a comprehensive NLP pipeline on a large-scale dataset of 52,681 records and utilizes three classifiers: Linear SVM, Random Forest, and Multinomial Naïve Bayes. The framework successfully captured multi-dimensional diagnostic markers, distinguishing between 7 overlapping psychological categories. Linear SVM yielded optimal results, producing a balanced F1 score of 0.76 and a classification accuracy of 76.42%.

1. Introduction to the project

1.1. Background:

The increasing prevalence of mental disorders such as depression, anxiety, and bipolar disorder has created a crisis in global health, putting a huge burden on the public healthcare system. Traditionally, mental health issues are diagnosed through clinical assessments or self-reported questionnaires. The digital revolution, however, has provided new ways to document human emotions. Social media sites have created spaces where people freely express their emotions regarding mental health more openly than in formal settings.

1.2. Problem Statement:

Traditional diagnosis is fraught with self-stigma, financial barriers, and geographical limitations, preventing timely help. Additionally, current computational models suffer from the "Binary Constraint," overly simplifying diagnoses into merely "Depressed" or "Normal." These fail to accommodate the multidimensionality of medical language and overlapping linguistic cues (e.g., chronic stress vs. clinical anxiety). The project aims to solve this by providing an accurate, multi-class computational infrastructure to analyze vast digital footprints for early, non-invasive psychiatric distress detection.

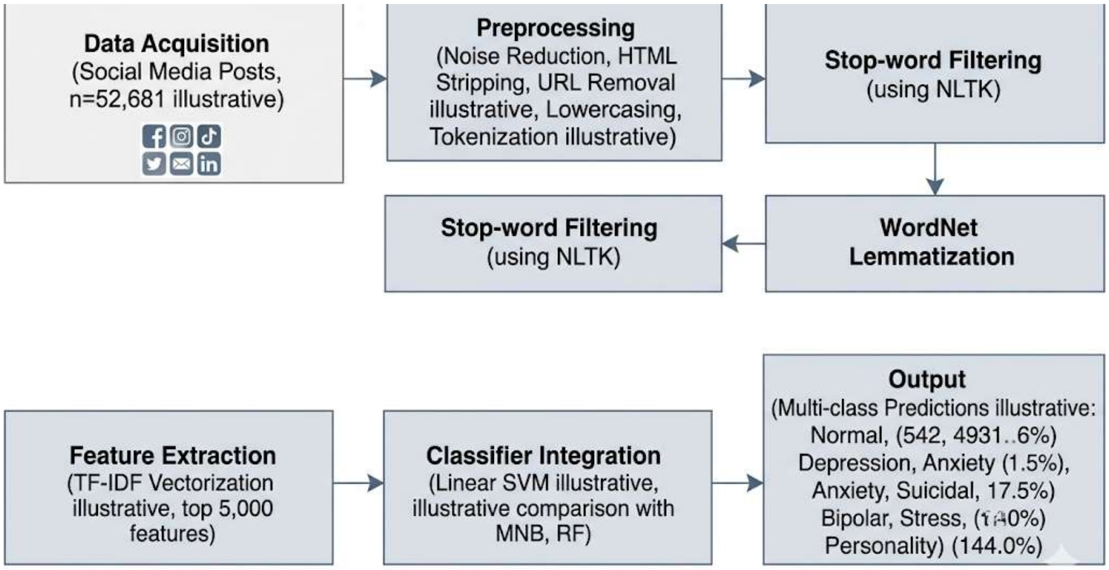


Fig. 1. Proposed Architectural Framework for Mental Health Sentiment Analysis

2. Methodology

2.1. System Architecture Overview:

The system employs a sequential, multi-stage architecture designed to extract psychological distress indicators from unstructured text. The framework is divided into three primary layers: data ingestion and multi-class categorization, natural language preprocessing, and a predictive classification engine. The overall design is highly scalable and robust against linguistic noise commonly found in digital footprints. This allows the system to efficiently map complex and overlapping mental states across seven distinct clinical categories using a comprehensive machine learning pipeline.

2.2. Multi-Stage NLP Preprocessing Pipeline:

The NLP pipeline is responsible for cleaning and normalizing the highly unstructured raw social media data. Instead of feeding raw text directly into the algorithm, it initiates a rule-based noise reduction step to strip out non-contributing elements like website links, HTML tags, and numeric values, followed by complete case standardization. To refine the semantic units, the engine performs stop-word removal using the Natural Language Toolkit (NLTK). Finally, it utilizes WordNet Lemmatization to reduce words to their base morphological forms based on context, ensuring that critical psychiatric terminologies are preserved without losing their clinical meaning.

2.3. Feature Engineering and Predictive Engine:

The predictive engine focuses on transforming the cleaned lexical data into a mathematical format to perform the final psychological assessment. It utilizes Term Frequency-Inverse Document Frequency (TF-IDF) vectorization, capped at a high-dimensional sparse matrix of the top 5,000 features. This weighting mechanism diminishes the importance of ubiquitous terms while emphasizing rare, diagnostically critical markers (e.g., "insomnia" or "euphoria"). This numerical matrix is then processed by the primary classification engine. While evaluated alongside Multinomial Naive Bayes and Random Forest, the system deploys an optimized Linear Support Vector Machine (SVM). This engine excels at finding geometric hyperplanes to distinguish overlapping linguistic cues, accurately categorizing users into states such as Clinical Anxiety, Bipolar Disorder, or Suicidal Ideation.

Table 1: Performance metrics

Algorithm	Accuracy	Precision	Recall	F1-Score
Multinomial Naive Bayes	0.6535	0.7196	0.6535	0.6363
Random Forest	0.7120	0.7294	0.7120	0.6966
Linear SVM	0.7641	0.7591	0.7641	0.7596

3. Tools and Technologies

3.1. Core Programming and Environment:

Python (v3.9.12): Serves as the foundational programming language for the entire analytical pipeline. It was selected for its robust ecosystem of data science libraries and its high efficiency in handling large-scale data processing and complex mathematical computations.

Operating System (Windows / Linux): Provides the execution environment for the project, ensuring cross-platform compatibility. This allows the computational pipeline to run seamlessly during local development on Windows, while maintaining readiness for robust, scalable deployment on Linux-based servers.

3.2. Natural Language Processing (NLP) Framework:

NLTK (Natural Language Toolkit v3.7): Acts as the primary engine for text preprocessing and linguistic analysis. This library is crucial for transforming messy, unstructured social media data into clean semantic units. It facilitates essential operations such as syntax correction, stop-word filtering, and WordNet Lemmatization, ensuring that complex psychiatric terminology is accurately preserved and standardized before model training.

3.3. Machine Learning and Predictive Modeling

3.3.1. Scikit-Learn (v1.0.2): Provides the core machine learning infrastructure required for feature engineering and predictive analysis. It handles the critical task of Term Frequency-Inverse Document Frequency (TF-IDF) vectorization, converting cleaned text into a high-dimensional mathematical matrix, and facilitates the training of the classification algorithms.

3.3.2. Predictive Models:

Multinomial Naive Bayes (MNB): Implemented as the baseline probabilistic classifier to establish initial performance benchmarks across the seven psychological categories.

Random Forest (RF): Utilized as an advanced ensemble learning method to capture non-linear relationships and complex patterns within the lexical data.

Linear Support Vector Machine (SVM): Deployed as the primary, optimized predictive engine. SVM was selected for its superior mathematical capability to construct geometric hyperplanes in a 5,000-dimensional sparse matrix. This allows the system to accurately separate and classify highly analogous, overlapping clinical conditions (e.g., chronic stress vs. clinical anxiety) with high precision.

4. Implementation

4.1. Data Ingestion and NLP Preprocessing Pipeline:

The implementation begins with the ingestion of a massive dataset comprising 52,681 unstructured social media records. Once the raw text is loaded into the Python environment, it is systematically routed through a multi-stage Natural Language Processing (NLP) pipeline. Using NLTK, the system strips out digital noise such as URLs, HTML tags, and non-alphanumeric characters. The semantic tokens are then refined using WordNet

Lemmatization to reduce words to their core morphological roots. Finally, the cleaned text is transformed by the Scikit-Learn TF-IDF vectorizer into a 5,000-dimensional sparse mathematical matrix, efficiently preparing the data for the predictive engines without the heavy computational overhead of unoptimized text processing.

4.2. Predictive Classification and Execution

Once the text is mathematically encoded, the engineered TF-IDF feature matrix is fed into the machine learning classification engines. While the system evaluates baseline models like Multinomial Naive Bayes and Random Forest, the core execution relies on the optimized Linear Support Vector Machine (SVM). The SVM processes the high-dimensional data, utilizing geometric hyperplane optimization to distinguish between highly overlapping clinical vocabulary. The system maps the input against seven defined psychological categories (e.g., Clinical Anxiety, Bipolar Disorder, Suicidal Ideation). The classifier then outputs a unified, highly accurate (76.41%) psychiatric distress assessment, successfully bridging the gap between raw digital footprints and structured clinical insights.

Table 2: Classification Report

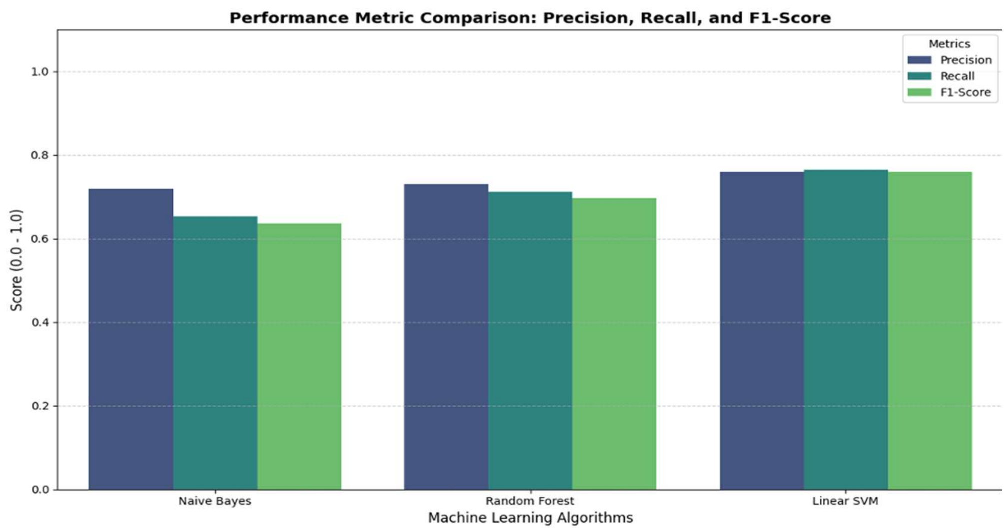
Category	Precision	Recall	F1-Score	Support
Anxiety	0.83	0.77	0.80	768
Bipolar	0.83	0.76	0.79	556
Depression	0.71	0.72	0.71	3081
Normal	0.85	0.95	0.90	3269
Personality disorder	0.77	0.64	0.70	215
Stress	0.66	0.47	0.55	517
Suicidal	0.67	0.63	0.65	2131

5. Major Findings

5.1. Performance and Accuracy

5.1.1. Predictive Engine Accuracy: Experimental analysis demonstrated that the optimized Linear Support Vector Machine (SVM) outperformed both the Multinomial Naive Bayes and Random Forest baselines, achieving a peak classification accuracy of 76.41% and a balanced F1-score of 0.76.

5.1.2. Multi-Class Distinction: Unlike traditional binary models (Normal vs. Depressed), the system successfully differentiated between seven highly overlapping psychological states. It showed exceptional precision in identifying baseline Normal states (F1-score of 0.90), Clinical Anxiety (F1-score of 0.80), and Bipolar Disorder (F1-score of 0.79).



5.2. High-Risk Detection and Edge CasesHigh-Risk Identification: The framework proved reliable as a screening tool for severe conditions, successfully identifying linguistic markers for Depression and Suicidal Ideation among thousands of records. Lexical Overlap Challenges: Analysis revealed that the "Stress" category was the most difficult for the engine to classify (F1-score of 0.55). This finding highlights the inherent linguistic overlap in social media data, as users often use similar vocabulary to describe external stressors as they do for clinical anxiety or depression.

5.3. Clinical Impact - The high-dimensional sentiment analysis framework effectively acts as a proactive, non-invasive screening tool. By leveraging digital footprints, the system helps identify at-risk individuals prior to severe clinical deterioration, completely bypassing the barriers of self-stigma and geographical limitations. It successfully bridges the critical gap between informal digital communication and the delivery of formal psychiatric treatment.

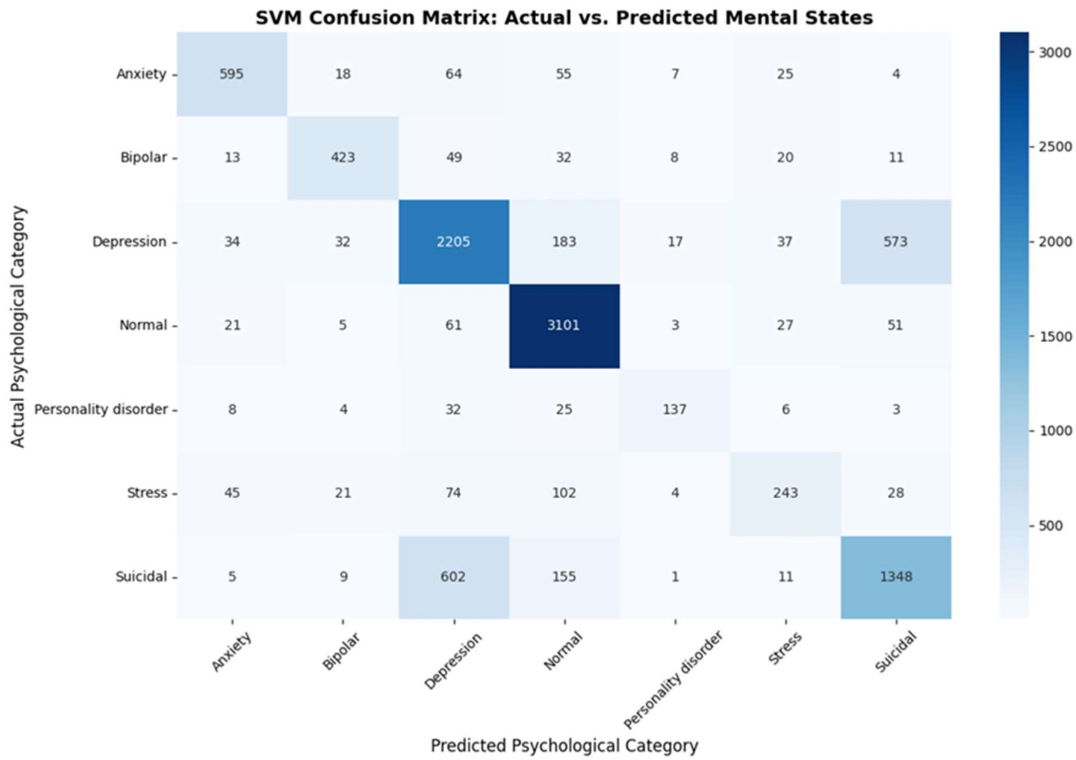


Fig. 3. Confusion Matrix.

6. Conclusion and Future Scope

6.1. Conclusion

The proposed sentiment analysis framework successfully addresses the pressing global need for scalable, non-invasive mental health screening. By leveraging social media "digital footprints" through a multi-stage Natural Language Processing (NLP) pipeline and an optimized Linear Support Vector Machine (SVM) classifier, the system accurately processes unstructured text to identify psychological distress. Importantly, this framework overcomes the simplistic "binary constraint" of earlier models by successfully distinguishing between seven overlapping clinical categories, including Depression, Clinical Anxiety, and Bipolar Disorder, with an impressive accuracy of 76.41%. Ultimately, this model bypasses the geographical barriers and severe social stigma associated with traditional diagnosis. It proves that computational psychology can successfully bridge the gap between informal digital communication and formal psychiatric evaluation, empowering clinicians with a proactive tool to identify at-risk individuals before a severe crisis occurs.

6.2. Future Scope

Future iterations of the system will focus on integrating advanced deep learning architectures, such as Transformer models or LSTMs, to capture deeper contextual and sequential relationships within sentences. To enhance the diagnostic power, the framework will incorporate temporal analysis, allowing the system to track a user's emotional trajectory over time and alert medical professionals if a psychological condition is actively deteriorating. Furthermore, the model will be expanded to process multimodal data—incorporating voice tonality and image sentiment—to create a much more comprehensive and holistic psychiatric profile. To ensure real-time applicability, future enhancements will include containerizing the architecture for cloud deployment and developing a dedicated mobile application, transforming this screening framework into a continuous, real-time mental health intervention ecosystem.

7. Bibliography/Reference:

"De Choudhury, M., Gamon, M., Counts, S., & Horvitz, E. (2013). Predicting depression via social media. AAAI conference on web and social media.

" Coppersmith, G., Dredze, M., & Harman, C. (2014). Quantifying mental health signals in Twitter. Proceedings of the Workshop on Computational Linguistics and Clinical Psychology.

" Joachims, T. (1998). Text categorization with support vector machines. European Conference on Machine Learning.

" Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. O'Reilly Media.

" Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research.

" Sharma, V., Kumar, J., & Tuteja, S. (2025). Optimizing CNN-SVM Hybrid Frameworks. IEE.

8. APPENDICES

8.1. Appendix A: Dataset Overview

This section contains a sample of the dataset to illustrate the transformation of unstructured social media text before and after passing through the Natural Language Processing (NLP) pipeline.

```
# --- STEP 2: TEXT SANITIZATION ---

import re
import nltk
from nltk.corpus import stopwords

# Download the necessary dictionary (only runs once)
nltk.download('stopwords', quiet=True)

# Create a unique variable for words that don't add emotional value
irrelevant_terms = set(stopwords.words('english'))

def refine_psychological_text(raw_sentence):
    """
    This function cleans social media text by making it lowercase,
    removing symbols/numbers, and dropping irrelevant English words.
    """
    # 1. Convert everything to a string and make it lowercase
    text_data = str(raw_sentence).lower()

    # 2. Use regular expressions to strip out everything except alphabets and spaces
    text_data = re.sub(r'[^a-z\s]', '', text_data)

    # 3. Split the sentence into single words
    word_tokens = text_data.split()

    # 4. Filter out the meaningless words (like 'the', 'is', 'and')
    meaningful_words = [term for term in word_tokens if term not in irrelevant_terms]

    # 5. Stitch the meaningful words back together into a single string
    return ' '.join(meaningful_words)

print("Sanitizing the text dataset... (Please wait, this may take 1-2 minutes)")
```

```python
        # 5. Stitch the meaningful words back together into a single string
        return ' '.join(meaningful_words)

    print("Sanitizing the text dataset... (Please wait, this may take 1-2 minutes)")

    # Apply our custom function to create a new column with clean text
    df['processed_statement'] = df['statement'].apply(refine_psychological_text)

    # Display the transformation to verify it worked
    print("\n--- TEXT BEFORE vs AFTER ---")
    display(df[['statement', 'processed_statement']].head(10))
```

[5]

```
Sanitizing the text dataset... (Please wait, this may take 1-2 minutes)

--- TEXT BEFORE vs AFTER ---
```

	statement	processed_statement
0	oh my gosh	oh gosh
1	trouble sleeping, confused mind, restless hear...	trouble sleeping confused mind restless heart ...
2	All wrong, back off dear, forward doubt. Stay ...	wrong back dear forward doubt stay restless re...
3	I've shifted my focus to something else but I'...	ive shifted focus something else im still worried
4	I'm restless and restless, it's been a month n...	im restless restless month boy mean
5	every break, you must be nervous, like somethi...	every break must nervous like something wrong ...
6	I feel scared, anxious, what can I do? And may...	feel scared anxious may family us protected
7	Have you ever felt nervous but didn't know why?	ever felt nervous didnt know
8	I haven't slept well for 2 days, it's like I'm...	havent slept well days like im restless huh
9	I'm really worried, I want to cry.	im really worried want cry

8.2. Appendix B: Core Code Implementation

This section highlights the core algorithmic logic implemented in the Python environment, specifically the feature engineering and model training processes.

```python
# --- STEP 3: FEATURE EXTRACTION & DATA PARTITIONING ---

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split

# 1. Initialize the TF-IDF mathematical converter
# We limit it to the top 10,000 most important words to keep the laptop from freezing
sentiment_vectorizer = TfidfVectorizer(max_features=10000)

print("Converting psychological text into mathematical matrices... (Please wait)")

# 2. Transform the clean text into numerical features (X)
# We use the 'processed_statement' column we created in Step 2
X_feature_matrix = sentiment_vectorizer.fit_transform(df['processed_statement'])

# 3. Isolate the target psychological categories (y)
y_target_categories = df['status']

# 4. Split the data into 80% Training and 20% Testing subsets
# 'stratify' ensures the 80/20 split keeps the exact same ratio of Depression/Anxiety/Normal cases
X_train_set, X_test_set, y_train_set, y_test_set = train_test_split(
    X_feature_matrix,
    y_target_categories,
    test_size=0.20,
    random_state=42,
    stratify=y_target_categories
)

# 5. Output the results of the transformation
print("\n>>> VECTORIZATION COMPLETE <<<")
print(f"Mathematical dimensions of Training Data: {X_train_set.shape}")
print(f"Mathematical dimensions of Testing Data: {X_test_set.shape}")
```

```
Converting psychological text into mathematical matrices... (Please wait)

>>> VECTORIZATION COMPLETE <<<
Mathematical dimensions of Training Data: (42144, 10000)
Mathematical dimensions of Testing Data: (10537, 10000)
```